Overview

- This week's exercises are roughly divided into two sections:
  - First three exercises are mainly based around the **joystick**
  - Last three are focused on the **Encoder and ISR** and do not require the use of a joystick
  - You might want to keep these codes separate, so that you won't have unnecessary code pieces confusing you

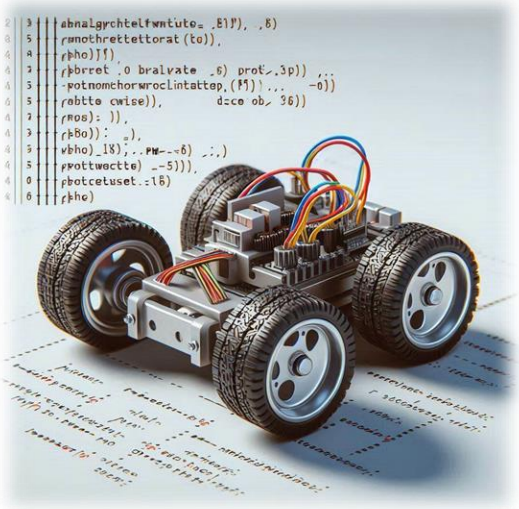
- By the end of these exercises, you should be (separately) able to:
  - drive your car with the joystick
  - tell the car to go forwards until it has driven a set distance

AnalogWrite and Pulse Width Modulation (PWM)

Exercise 1:

- Test out how to drive the motors with the analogWrite()- and digitalWrite()-functions
  - Use the provided motor test-code to your advantage
  - What changes the speed? How about direction?
  - What is the maximum value for speed?

- Make a simple code, where your car does these steps once:
  - Rotate left wheel forward, 50 % of maximum speed, duration 1 second
  - Rotate right wheel forward, 50 % of maximum speed, duration 1 second
  - Rotate both wheels backwards, 25 % of maximum speed, duration 2 seconds
  - Stop

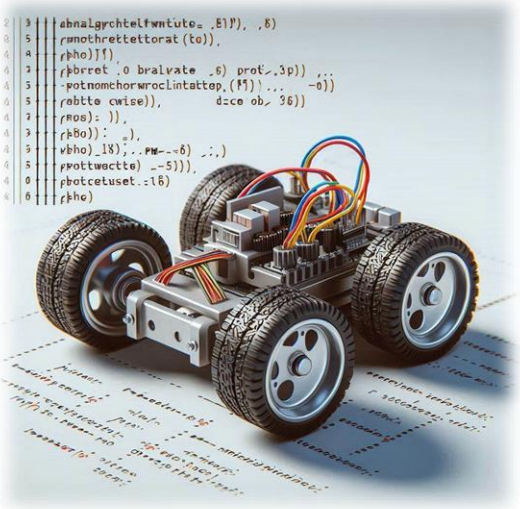


AnalogWrite and PWM recap video click

AnalogWrite and Pulse Width Modulation (PWM)

Exercise 2:

- Use the joystick and its analog values (x- and y-values) to drive your car.
  - determine which one (x/y) is used for your car's speed, and which is used for turning/direction
  - consider what should happen if the joystick is not used (note the values) and what determines if we want the car to move backwards/forwards



- Optimize vehicle response to joystick, the car should:
  - move forwards/backwards and turn left and right
  - move faster or slower depending on the joystick values
  - not move when joystick isn't used

Competition 1 - Speed

Exercise 3:

- Prepare your code for a speed competition
- The competition’s target is to drive as fast as possible a simple track using the joystick to navigate your car from start to finish
- Fastest team gets an extra point, so be on time for the lessons when these exercises have their deadline (DL)
- Competitions are only for that one day, no late submissions



ISR and Encoder Pulses

Exercise 4:

- Create a program that drives your car forwards for 3 seconds and simultaneously counts the amount of Encoder pulses from pin ENCA (right or left) using the ISR (attachInterrupt). Measure/note the following things:
  - How many pulses do you get for the given time period?
  - What was the distance driven in centimeters?
  - How many pulses per centimeter?
  - Is the pulse count same when driving on the table vs floor?

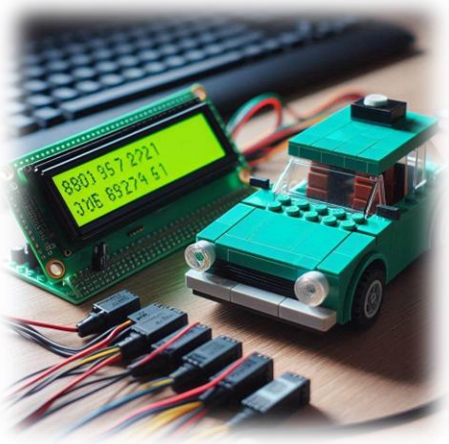


See wiring help in file: Encoder wiring

ISR and Encoder Pulses

Exercise 5:

- Create a program that allows you to drive a set distance, for example 10cm, forwards
- You can hard code a variable to set the distance in the beginning of your code
- Use the pulses per cm information from last exercise to your advantage
- Print both *ENC A left* and *ENC A right* pulse counts on to the LCD



Competition 2 - Accuracy

Exercise 6:

- Prepare your code for an accuracy competition
- The competition’s aim is to drive your vehicle an undetermined distance forwards by using the encoder pulses to determine the distance driven
- Closest team to the set distance gets an extra point, so be on time for the lessons when these exercises have their deadline
- Competitions are only for that one day, no late submissions

